MetaboTyping RA Agent Workbench

Final capability and data-landscape report

Condensed evidence report | MW corpus snapshot: 24 August 2026, 04:29 UTC

This report answers two questions: what is present in the current Metabolomics Workbench (MW) bulk indexes, and what can the MetaboTyping RA Agent Workbench do with repository-scale metabolomics evidence? Counts are frozen to one provenance-tracked retrieval. Every number retains its denominator; study summaries, analyses, organism incidences, reported feature counts, and RefMet nomenclature entries are intentionally kept separate.

Executive measure	Audited result	Interpretation
MW study summaries	4,499	Unique ST records after one additional row sharing a stud...
MW analyses	7,367	Unique analysis IDs after three additional rows sharing a...
Organism coverage	4,483 studies; 4,770 incidences; 481 labels	4,395 single-species and 88 multi-species studies
Assay evidence	2,385 untargeted; 14 explicit targeted; 4,968 unclassifi...	Analysis-level categories; targeted is a strict lower bou...
Reported feature volume	1,231,376 instances across 5,420 numeric records	Median 99 per reported analysis; not unique metabolites
RefMet reference snapshot	205,948 entries	23 superclasses, 333 main classes, 960 subclasses
Workbench inventory	22 canonical roles; 24 logical skills	Two deprecated role aliases are not extra capabilities
Source-routing inventory	28 sources; 27 lanes	Registration does not imply that every source has a live ...

1. Metabolomics Workbench corpus

Study and species landscape

The official MW bulk endpoints yielded 4,500 summary rows and 7,370 analysis rows. Deterministic de-duplication retained the first record per identifier and removed one additional row sharing study ID ST002097 plus three additional rows sharing analysis IDs, leaving 4,499 studies and 7,367 analyses. This rule establishes identifier uniqueness, not that the removed rows were content-identical. The species endpoint contained 4,773 rows and 4,770 unique study-Latin-name pairs. It covered 4,483 study IDs: 4,395 listed one species and 88 listed more than one. Eighteen summary studies had no species-index row, while two species-index IDs had no matching summary row. Species percentages therefore use 4,483 species-annotated studies as the denominator and may sum above 100%.

Rank	MW source label	Studies	Share of annotated studies
1	Homo sapiens	1,852	41.31%
2	Mus musculus	1,463	32.63%
3	Rattus norvegicus	257	5.73%
4	Escherichia coli	54	1.20%
5	Plasmodium falciparum	53	1.18%
6	Drosophila melanogaster	32	0.71%
7	Caenorhabditis elegans	27	0.60%
8	Arabidopsis thaliana	25	0.56%
9	Danio rerio	24	0.54%
10	Saccharomyces cerevisiae	24	0.54%

Targeted, untargeted, and unresolved assay scope

MW exposes a dedicated untarg_studies index but no parallel targeted-only bulk index. That endpoint identifies 2,385 untargeted analyses in 1,250 studies. Searching analysis text conservatively identifies 14 explicitly targeted analyses in 14 studies. The other

4,968 analyses are unclassified or other; they cannot be relabeled targeted from the absence of an untargeted flag. At study level, 3,235 studies are neither explicitly targeted nor present in the untargeted index (or have no analysis row). This is an evidence gap, not a binary assay classification.

The number_of_metabolites endpoint supplied 5,485 records: 5,420 numeric values, 65 blanks, and no record for 1,882 analyses. Numeric values ranged from 1 to 11,992 (median 99) and summed to 1,231,376 analysis-feature instances. Because the same chemical can recur across analyses and an untargeted feature need not be an identified compound, this sum is not a unique-metabolite count.

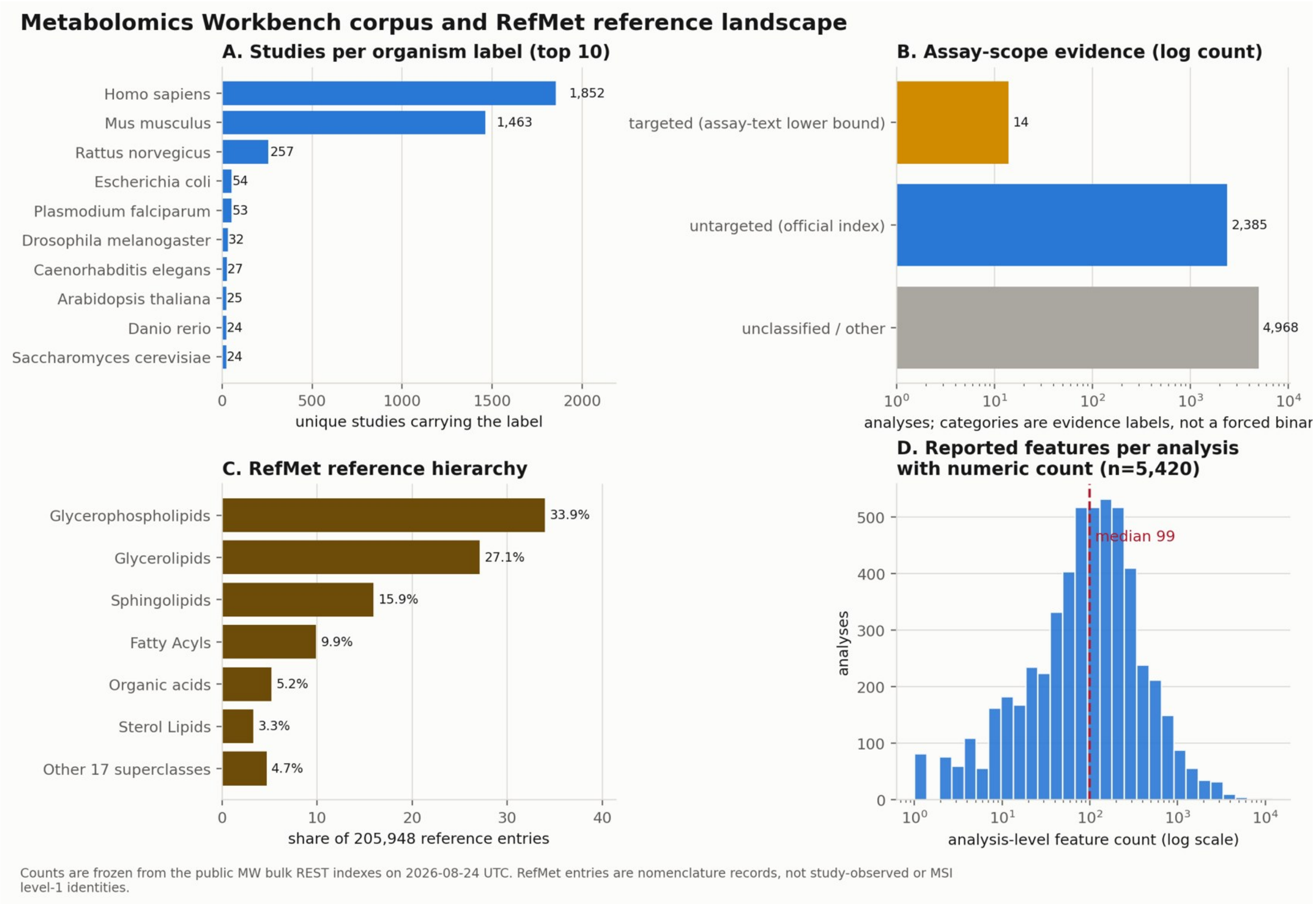

Figure 1. Frozen MW corpus landscape. Species use study incidences; assay categories use analyses; RefMet percentages describe a separate reference nomenclature snapshot; feature counts are analysis-level reported values.

2. RefMet chemical hierarchy

The requested superclass/class/subclass distribution is implemented as the local RefMet hierarchy (super_class, main_class, sub_class), not a PubMed or PubChem classification. The snapshot contains 205,948 unique reference nomenclature entries, 205,919 with a nonblank formula, spanning 23 superclasses, 333 main classes, and 960 subclasses. Its release identifier was not recorded in the local source file. These are possible reference labels-not 205,948 chemicals demonstrated in MW-and a shared formula does not establish identity.

Rank	Superclass (n)	Main class (n)	Subclass (n)
1	Glycerophospholipids (69,886)	Triradylglycerols (33,688)	TAG (33,431)
2	Glycerolipids (55,850)	Glycosphingolipids (16,592)	FAHFA (14,719)
3	Sphingolipids (32,811)	Fatty esters (15,818)	HexCer (10,579)
4	Fatty Acyls (20,340)	Glycerophosphoglycerols (15,387)	Tripeptides (8,017)
5	Organic acids (10,648)	Glycerophosphoethanolamines (12,424)	O-DG (5,770)

The first three superclasses account for 76.98% of entries, showing that the reference file is lipid-heavy. That composition describes the nomenclature resource; it must not be projected onto MW study observations without a study-analysis-feature crosswalk and identity evidence.

3. Demonstrations: MoTrPAC comparison and Lac-Phe

Acute-human MW-MoTrPAC comparison

The cached acute-human comparison uses MW study ST004303 (266 effect rows; 261 case/space-normalized labels) and the cached public MoTrPAC human acute-endurance late-post-versus-pre plasma effect table (1,657 rows; 1,509 normalized labels across 11 targeted and untargeted panels). It finds 99 exact normalized-label overlaps and 102 punctuation-insensitive candidate key overlaps. Duplicate labels or panels make 46 candidates ambiguous; 56 are unique on both sides at the duplicate-key screening gate. Neither effect table carries source-stable RefMet identifiers, so zero candidates are declared structurally validated or poolable. The result is a review queue, not a meta-analysis.

Lac-Phe worked example

The exact N-Lactoyl phenylalanine RefMet-name query returned 85 raw MW metabolite-statistics endpoint rows; two additional name variants returned none. After de-duplication it retained 60 study-analysis pairs from 57 studies, including 24 analyses from 22 human studies. The resolver annotates N-Lactoyl phenylalanine as RefMet RM0131640 (formula $C_{12}H_{15}NO_4$; PubChem 11075454), but the inspected public source did not document linked authentic-standard retention-time plus MS/MS evidence for the focal feature. Identity therefore remains human-review required rather than being promoted to MSI level 1.

For the study-reported whole-blood feature in ST003662, 116 of 137 participant rows had complete positive-abundance pre/post values; 92 of 116 increased. The acute post-minus-pre paired log2 analysis estimated mean log2 fold change +1.1586 (geometric mean paired fold 2.23x), $p = 2.73e-15$, BH FDR = 2.50e-14-the fourth-largest log2 fold change among 165 usable all-athlete estimates. Arithmetic means rose from 0.539 to 0.755 micromol/L (ratio 1.40); this ratio and the paired log-scale estimand should not be interchanged. Protocol modality, intensity, duration, and post-exercise sampling delay were not encoded in the inspected source.

Across the RefMet snapshot, 22 labels begin with N-Lactoyl; 17 have at least one MW metabolite-statistics label association. Only the Lac-Phe label is associated with ST003662, and none of the 22 labels appears in the inspected public MoTrPAC human or rat tables. The workbench records this as a coverage gap, not evidence of biological absence.

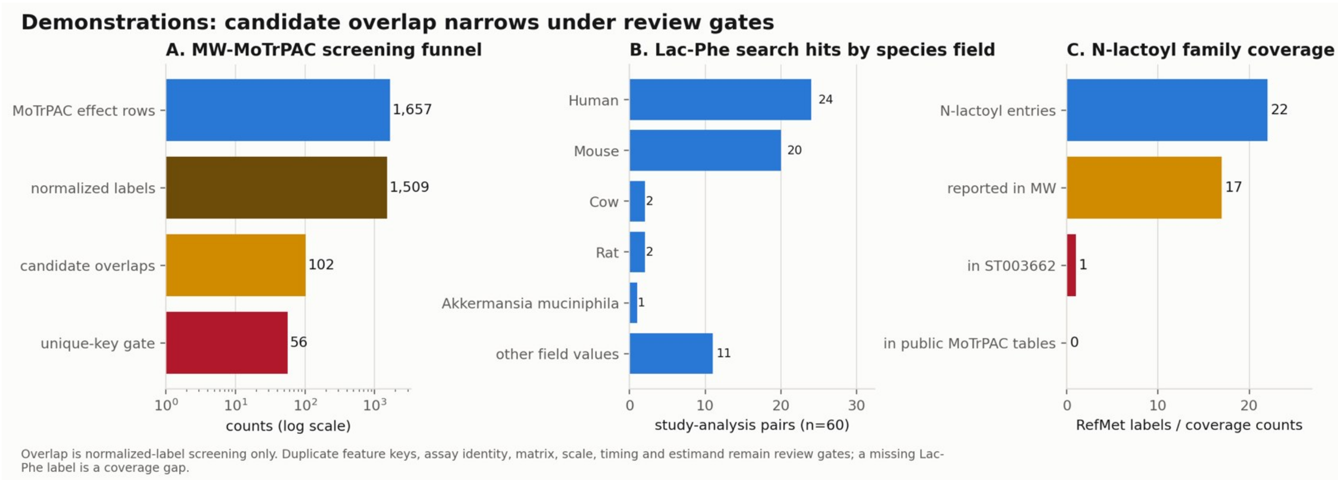

Figure 2. Demonstration funnels. Name-key overlap narrows to 56 unambiguous candidates but not verified identity; the Lac-Phe RefMet label is associated with several MW-reported species fields; N-lactoyl coverage differs across RefMet, MW, the focal study, and cached MoTrPAC tables.

4. What the workbench can do

The system combines declarative scientific roles and skills with deterministic Python workflows. Codex and Claude carry paired copies of the same 24 logical project skills, so those files are counted once. Its 22 canonical agent roles cover orchestration, literature and repository retrieval, metadata extraction and criticism, population/protocol review, metabolite and assay identity, variable crosswalks, estimand and synthesis review, readiness scoring, MoTrPAC alignment, visualization, and pipeline planning.

Workflow stage	Reviewable output	Fail-closed scientific behavior
Define and discover	Inclusion criteria, ranked studies, literature and reposit...	Missing repository support becomes a mirage/escalation flag
Extract and preserve	Study, dataset, variable, timing, assay, and provenance ca...	Unknown is never silently converted to absent
Resolve and harmonize	Identity candidates, confidence-scored crosswalks, assay-a...	Similar names alone never authorize a merge or pooling
Compare and visualize	MoTrPAC-style effects, overlap screens, descriptive RefMet...	Descriptive overlap is not called replication or pathway in...
Review and reproduce	Queues, advisory domain packets, manifests, scored readine...	Domain packets remain advisory; review-required mappings em...

The registry describes 28 sources across 27 retrieval lanes. It includes live core adapters, offline snapshots and fixtures, optional life-science connectors, standards references, and planned/manual lanes. Registration is routing metadata, not proof of live retrieval. The present synthetic/local release gate passes 148 of 148 tests. In its synthetic benchmark, candidate precision/recall and review capture are 1.00, unsafe auto-accept is 0/12, and quality-score agreement within 0.15 is 8/10. These are reproducibility checks on constructed fixtures, not independent external validation.

Current boundaries are deliberate: no automatic structure-level reconciliation across all sources; no inference that an unreported feature is biologically absent; no pooling across incompatible matrices, assays, units, scales, time points, protocols, or estimands; and no claim that all registered sources have executable adapters. The report itself is reproducible from the frozen MW manifest, derived evidence bundle, skill audit, scientific-readiness audit, and full Lac-Phe dossier.

Reusable two-paragraph summary

The MetaboTyping RA Agent Workbench is a provenance-first, human-in-the-loop system for metabolomics discovery, curation, harmonization, and comparison. The current worktree defines 22 canonical agent roles and 24 logical project skills paired across Codex and Claude; two additional role names are deprecated aliases, not distinct capabilities. It turns a research question into explicit inclusion criteria, discovers and ranks studies, preserves one-to-many repository records, extracts schema-validated study/dataset/variable cards, resolves or escalates metabolite identity, builds confidence-scored crosswalks and assay-aware harmonization plans, assesses metadata-backed readiness, appraises literature, and produces MoTrPAC-style plots, review packets, provenance manifests, and reproducible reports. Deterministic rules fail closed: a shared name never authorizes a merge, uncertain mappings remain under human review, and incompatible assays, matrices, units, scales, protocols, or estimands are not authorized for pooling without compatibility evidence and human adjudication.

At repository scale, the frozen MW indexes contain 4,499 study summaries and 7,367 analyses; 4,483 study IDs carry 4,770 study-organism incidences spanning 481 source labels. The official untargeted index covers 2,385 analyses in 1,250 studies, while only 14 analyses are explicitly targeted in assay text and 4,968 remain unclassified, so that remainder cannot be relabeled targeted. Separately, the local RefMet snapshot contains 205,948 reference entries across 23 superclasses, 333 main classes, and 960 subclasses-not 205,948 metabolites proven observed in MW. In the cached acute-human comparison, MW ST004303 and the MoTrPAC acute-endurance plasma table yield 102 punctuation-insensitive candidate name-key overlaps, only 56 unambiguous at the duplicate-key screen. The Lac-Phe RefMet label is associated by MW metabolite statistics with 57 studies and 60 study-analysis pairs; the study-reported Lac-Phe feature in ST003662 rose by mean paired log2 fold change +1.16 across 116 complete pairs, but no Lac-Phe label appears in the inspected public MoTrPAC tables. The workbench reports those coverage and identity gaps and routes them to human review rather than claiming null effects, replication, or poolable chemistry.

Provenance note. MW counts were retrieved through the repository's network-allowlisted fetcher from documented MW REST v1.2 bulk endpoints and frozen with raw payloads, normalized tables, SHA-256 hashes, endpoint URLs, timestamps, row counts, and de-duplication rules. RefMet, Lac-Phe, MoTrPAC, agent/skill, and test counts are local snapshot results; their source files and derived tables are enumerated in data/report_metrics.json and the adjacent CSV bundle.